# Marius Arcay

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## Education

**San Jose State University** — **San Jose, CA**
*Bachelor of Science in Computer Science* — Spring 2026 – Spring 2028 (Expected)

**West Valley College** — **Saratoga, CA**
*Associate of Science in Physics* — Winter 2026 – Spring 2028 (Expected)
*Associate of Science in Mathematics* — Fall 2022 – Winter 2026

## Experience

**Contract 3D Artist** — Oct 2025 – Jan 2026
*Valve Corporation* — *Remote*

- Authored **Rooftop**, an official Counter-Strike 2 multiplayer map using Valve's proprietary technology, balancing visual fidelity, gameplay, and performance
- Built, profiled and optimized scene geometry, materials, lighting, and assets to meet performance standards expected of an official Counter-Strike 2 map
- Shipped to a global audience of millions of players

**Retail Tech Consultant** — Jun 2021 – Present
*Target* — *Morgan Hill, CA*

- Provided personalized expertise to help guests select, understand, and purchase electronics

## Projects

**CUDA Fractal Renderer** [Technical Write-up] | *C++, CUDA, Multi-threading, Git, Visual Studio*

- Built a CUDA fractal renderer achieving a 100× GPU speedup over single-threaded CPU benchmarks
- Developed a benchmarking system comparing CPU/GPU speeds with dynamic iteration scaling and animation rendering
- Automated build setups to check for CUDA support and drop unneeded dependencies

**OpenGL Path Tracing Engine** [Technical Write-up] | *C++, OpenGL, GLSL, Git, Visual Studio*

- Built a GPU path tracer using OpenGL/GLSL with Monte Carlo sampling and Cook-Torrance GGX PBR maps
- Implemented a depth prepass to reduce overdraw alongside texture pooling and bounding-volume intersection tests
- Developed interactive rendering infrastructure featuring a translation gizmo, material editor, and debug modes

**3D Graphics Library** | *C++, Multi-threading, Visual Studio*

- Built a multi-threaded CPU 3D renderer in C++ featuring a model conversion pipeline and scene configuration

**Environment Monitoring Device** | *C, MySQL, I2C*

- Developed a Raspberry Pi device in C using I2C protocols to collect climate metrics and log data to MySQL

**Game Server Infrastructure** | *Python, Linux, OCI*

- Hosted Counter-Strike game servers on Oracle Cloud Infrastructure (OCI) Linux virtual machines
- Wrote a Python Discord bot using a game query library to keep track of server status and live player activity

## Technical Skills

**Languages**: C++, C, GLSL, Python, Java, Assembly (ARM)
**Graphics & GPU**: OpenGL, CUDA, PBR, Monte Carlo Rendering
**Systems & Cloud**: Linux (CLI/Bash), Oracle Cloud Infrastructure (OCI), Multithreading, I2C, Windows
**Libraries & Tools**: OpenCV, Raylib, Git, Visual Studio, VS Code